

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and

network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds

using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid

topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is

connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach

Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and	Good collaboration	Limited participation	No collaboration or contribution
Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		contribution (if team-based)			

Q1 – STAR TOPOLOGY FOR A 20-PERSON STARTUP

1. Problem Analysis

A startup with 20 employees requires a network that is simple, reliable, easy to manage, and expandable. The network must connect computers, printers, VoIP phones, laptops, and other devices.

The main requirements are:

- Connect all 20 employees.**
- Provide high-speed communication.**
- Allow easy troubleshooting.**
- Provide internet access.**
- Support wired and wireless devices.**
- Allow future expansion.**
- Minimize the effect of individual cable failures.**

A star topology is suitable because every device is connected to a central switch. If one workstation or its cable fails, the other devices can continue operating.

2. Solution Design

The proposed network consists of:

Computers/Printers/VoIP Phones → Managed Switch → Router/Firewall → Internet

A wireless access point is also connected to the switch for laptops and mobile devices.

A 24-port Layer 2/Layer 3 managed switch is recommended. It provides ports for the 20 employees and leaves additional ports for printers, access points, VoIP phones, and future devices.

3. Technical Implementation

Networking Devices

- 24-port managed switch**
- Router/firewall**
- Wireless access point**
- Computers**
- Printers**

- **VoIP phones**

Transmission Medium

Cat6 UTP cable is used between the switch and end devices.

Cat6 supports Gigabit-speed connections and can cover distances up to approximately 100 metres, which is suitable for a typical office floor.

Router/Firewall

The router/firewall provides:

- **Internet connectivity**
- **DHCP**
- **NAT**
- **Basic network security**

4. Problem-Solving Approach

The implementation follows these steps:

- 1. Identify the number of employees and devices.**
- 2. Select a 24-port managed switch.**
- 3. Install the switch at a central location.**
- 4. Connect all devices using Cat6 cables.**
- 5. Connect the switch to the router/firewall.**
- 6. Configure IP addresses and DHCP.**
- 7. Install the wireless access point.**
- 8. Test wired and wireless connectivity.**

5. Tools and Technologies

- **Managed Ethernet switch**
- **Router/firewall**
- **Cat6 UTP**
- **Wireless Access Point**
- **DHCP**
- **NAT**
- **Gigabit Ethernet**

6. Innovation

A useful improvement is to keep spare switch ports for future employees and devices. Centralized cable management also makes future moves, additions, and changes easier.

7. Testing and Validation

The following tests can be performed:

- Ping test between computers.**
- Internet connectivity test.**
- File-sharing test.**
- Printer connectivity test.**
- Wi-Fi coverage test.**
- Disconnect one workstation cable and verify that other users remain connected.**

The final test confirms the main advantage of star topology: failure of one workstation cable affects only that particular device.

8. Expected Result

The startup will obtain a reliable, easy-to-manage, scalable, and high-speed network. Individual device failures will not affect the entire network.

9. Conclusion

Star topology is the best choice for a 20-person startup because it provides centralized management, simple troubleshooting, easy expansion, and fault isolation.

Q2 – MESH TOPOLOGY FOR A BANK

1. Problem Analysis

A bank requires **near-zero downtime** because network failure can affect financial transactions and critical banking services.

The major requirements are:

- Very high reliability.
- Minimum downtime.
- Redundant network connections.
- High-speed communication.
- Protection against link failure.
- Low-latency communication.
- Reliable internet connectivity.

Therefore, a **full or partial mesh topology** is suitable for the bank's core network.

2. Solution Design

A **partial mesh topology** is recommended for practical deployment.

The core switches, routers, and firewalls are connected using multiple paths.

Distribution and access layers have redundant connections to at least two core switches.

Basic Design

Internet → Dual Routers → Core Switches → Distribution Switches → Access Switches → Users

The core devices have alternative paths so that traffic can continue even when one connection fails.

3. Technical Implementation

Networking Devices

- Two enterprise-grade routers
- Core switches
- Distribution switches
- Access switches
- Firewalls
- Servers
- UPS systems
- RAID storage

Dual routers can provide redundant internet connectivity using routing protocols such as **BGP**.

Transmission Medium

Fiber optic cables should be used between core devices.

- **Single-mode fiber:** Long-distance connections between buildings.
- **Multi-mode OM4:** Suitable for data-centre environments.

Fiber provides high bandwidth and is resistant to electromagnetic interference.

4. Problem-Solving Approach

1. Identify critical banking services.
2. Identify possible network failure points.
3. Design redundant core connections.
4. Install dual routers and firewalls.
5. Connect core devices using fiber.
6. Configure routing protocols.
7. Provide backup power.
8. Test link and device failures.
9. Monitor the network continuously.

5. Tools and Technologies

- Fiber optic cable
- Enterprise routers
- Core switches
- Firewalls
- BGP
- OSPF
- UPS
- RAID storage

Protocols such as **OSPF or BGP** can reroute traffic through alternate paths when a link fails.

6. Innovation

The main innovation is **redundancy at multiple levels:**

- Dual routers
- Multiple core links
- Redundant uplinks
- Backup power

- RAID storage

This reduces the possibility of a single point of failure.

7. Testing and Validation

Testing includes:

- Disconnecting one fiber link.
- Testing router failure.
- Testing core switch failure.
- Checking automatic traffic rerouting.
- Measuring network latency.
- Testing internet redundancy.
- Testing UPS backup operation.

The network should continue operating when an individual link fails.

8. Expected Result

The bank receives a **highly available and fault-tolerant network** capable of maintaining communication even during individual network failures.

9. Conclusion

Mesh topology is suitable for banking environments because redundant paths provide high availability, reliability, and rapid recovery from failures.

Q3.BUS TOPOLOGY FOR A SMALL CLASSROOM LAB

1. Problem Analysis

A small classroom laboratory has a **limited budget** and a relatively small number of computers. The network is mainly required for:

- Internet browsing.
- File sharing.
- Basic communication.
- Connecting student computers.
- Low-cost installation.

A traditional bus topology can reduce hardware requirements because computers share a common communication line.

2. Solution Design

The proposed design uses a common communication backbone.

Traditional Bus Design

Computer → T-Connector → Common Coaxial Cable → T-Connector → Computer

In the classic implementation:

- RG-58 50-ohm coaxial cable is used.
- BNC connectors connect computers.
- T-connectors attach devices to the backbone.
- 50-ohm terminators are placed at both ends.

3. Technical Implementation

Hardware

- RG-58 coaxial cable
- BNC connectors
- T-connectors
- 50-ohm terminators
- Computers

A modern low-cost alternative can use an inexpensive unmanaged switch with Cat5e/Cat6 patch cables, providing similar low-cost simplicity.

4. Problem-Solving Approach

1. Identify the number of computers.
2. Estimate the cable length.
3. Install the common communication backbone.
4. Connect each computer to the backbone.

5. Install terminators at both ends.
6. Configure computer network settings.
7. Test communication and file sharing.

5. Tools and Technologies

- Coaxial cable
- BNC connectors
- T-connectors
- Terminators
- Cat5e/Cat6 in modern variation
- Unmanaged switch
- Windows/Linux file sharing

No dedicated server is required; a teacher's computer can provide basic file-sharing services.

6. Innovation

The low-cost modern variation can use an inexpensive unmanaged switch and Cat5e/Cat6 cables instead of the traditional coaxial bus arrangement.

This reduces installation complexity while maintaining a low-cost network.

7. Testing and Validation

Testing includes:

- Ping between computers.
- Internet browsing test.
- File-sharing test.
- Cable connectivity test.
- Disconnecting a cable to observe the effect.
- Testing network performance with multiple computers.

8. Limitations

The major disadvantages are:

- Possible collisions.
- A backbone cable failure can affect all connected machines.
- Limited scalability.
- Troubleshooting can become difficult.

However, these limitations are acceptable for a supervised classroom with light network usage.

9. Expected Result

The classroom receives a **simple and inexpensive network** suitable for basic educational activities.

10. Conclusion

Bus topology is appropriate for a small classroom laboratory when cost is the primary concern and network traffic is relatively light. 4 –

Q4.HYBRID TOPOLOGY FOR A UNIVERSITY CAMPUS

1. Problem Analysis

A large university campus may have thousands of users distributed across:

- Lecture halls
- Laboratories
- Libraries
- Hostels
- Administrative buildings

A single topology cannot efficiently satisfy all these requirements.

The network requires:

- High bandwidth.
- Reliable communication.
- Inter-building connectivity.
- Wireless access.
- Redundancy.
- Easy maintenance.
- Future expansion.

Therefore, a **hybrid topology** is the most suitable solution.

2. Solution Design

The proposed network combines:

- **Ring/partial mesh** at the campus backbone.
- **Star topology** inside buildings.
- **Bus/tree structures** in selected laboratories.

This provides redundancy at the backbone and simple management inside buildings.

3. Technical Implementation

Core Layer

The campus core uses:

- High-speed routers.
- Multilayer switches.
- Fiber optic cables.
- Ring/partial mesh arrangement.

10 Gbps or 40 Gbps fiber connections can be used for the campus backbone.

Distribution Layer

Each building has:

- Distribution switch.
- Dual fiber uplinks to the campus core.

This provides redundancy if one uplink fails.

Access Layer

Access switches connect:

- Computers.
- Printers.
- IP devices.
- Wireless access points.

Cat6 cables connect individual devices to access switches.

4. Wireless Network

Wireless Access Points are installed across the campus.

They can receive power through **PoE**, allowing the same network cable to carry both power and data.

5. Problem-Solving Approach

1. Survey campus buildings.
2. Estimate users and devices.
3. Design the fiber backbone.

4. Install core routers and switches.
5. Connect buildings using dual fiber links.
6. Install distribution switches.
7. Install access switches.
8. Connect devices using Cat6.
9. Install PoE wireless access points.
10. Configure and test the network.

6. Tools and Technologies

- Fiber optic cable.
- Cat6 cable.
- Layer 2 switches.
- Layer 3 switches.
- Routers.
- Wireless Access Points.
- PoE.
- Core/distribution/access architecture.

7. Innovation

The design is highly scalable. When a new building is added, a new distribution switch can be installed and connected to the campus core without disrupting the existing network.

8. Testing and Validation

Testing includes:

- Fiber connectivity.
- Ping between buildings.
- Bandwidth testing.
- Wi-Fi coverage testing.
- PoE testing.
- Failure of one backbone link.
- Testing alternate paths.
- Load testing with multiple users.

9. Expected Result

The university gets:

- High-speed communication.
- Reliable connectivity.
- Better fault tolerance.

- Good Wi-Fi coverage.
- Easy troubleshooting.
- Simple expansion.
- Efficient campus-wide management.

10. Conclusion

A hybrid topology is the best solution for a large university because it combines the strengths of multiple topologies. It provides **redundancy at the backbone, manageability inside buildings, and scalability across the campus.**

Q 5 – NETWORK FOR MULTIPLE DEPARTMENTS

1. Problem Analysis

A company with departments such as **HR, Finance, Sales, and IT** needs a structured network that allows communication while maintaining security between departments.

The main problems are:

- Large number of users.
- Different departments have different requirements.
- Sensitive departmental information must be protected.
- Users need controlled communication.
- Network traffic must be efficiently managed.
- The company should be able to expand easily.

Therefore, a hierarchical network using **switches, routers, VLANs, and inter-VLAN routing** is suitable.

2. Solution Design

Each department receives a dedicated Layer 2 access switch.

Basic Architecture

Department Devices → Layer 2 Switch → Layer 3 Core Switch → Router/Firewall → Internet

Departments are separated using VLANs.

For example:

Department VLAN

HR	VLAN 10
Finance	VLAN 20
Sales	VLAN 30
IT	VLAN 40

This logical separation improves security and reduces unnecessary broadcast traffic.

3. Technical Implementation

Access Layer

Layer 2 switches connect:

- Computers.
- Printers.
- IP phones.
- Other departmental devices.

Core Layer

A Layer 3 switch performs **inter-VLAN routing**, allowing controlled communication between departments.

Router/Firewall

The router connects the internal network to the internet.

Firewall functionality can provide additional security.

4. IP Addressing and Subnetting

Each VLAN can have a separate IP subnet.

Example:

- HR → 192.168.10.0/24
- Finance → 192.168.20.0/24
- Sales → 192.168.30.0/24
- IT → 192.168.40.0/24

This makes network management and security easier.

5. Problem-Solving Approach

1. Identify all departments.
2. Count devices in each department.
3. Create separate VLANs.
4. Assign IP subnets.
5. Install Layer 2 access switches.
6. Connect them to the Layer 3 core switch.
7. Configure inter-VLAN routing.
8. Connect the core to the router/firewall.
9. Configure security policies.
10. Test connectivity and access restrictions.

6. Tools and Technologies

- Layer 2 switches.
- Layer 3 switch.
- Router.
- Firewall.
- VLAN.
- Inter-VLAN routing.
- IP addressing.
- Subnetting.
- Redundant links.

7. Innovation

A major improvement is using **department-based VLAN segmentation**.

For example, Finance computers can be isolated from unnecessary access by other departments while still allowing authorized communication.

Backup paths can also be included to improve reliability.

8. Testing and Validation

Testing should include:

- Ping within the same VLAN.
- Inter-VLAN communication test.
- Internet connectivity test.
- VLAN isolation test.

- Firewall access test.
- IP addressing test.
- Redundant-link failure test.
- Network performance test.

9. Expected Result

The company will have:

- Better security.
- Reduced broadcast traffic.
- Efficient communication.
- Centralized network management.
- Easier troubleshooting.
- Better scalability.
- Controlled inter-department communication.

10. Conclusion

A hierarchical network with **Layer 2 access switches, a Layer 3 core switch, VLANs, inter-VLAN routing, and a router/firewall** provides an efficient and secure solution for a multi-department company.